

DBsolo IKS automation and Jenkins Integration

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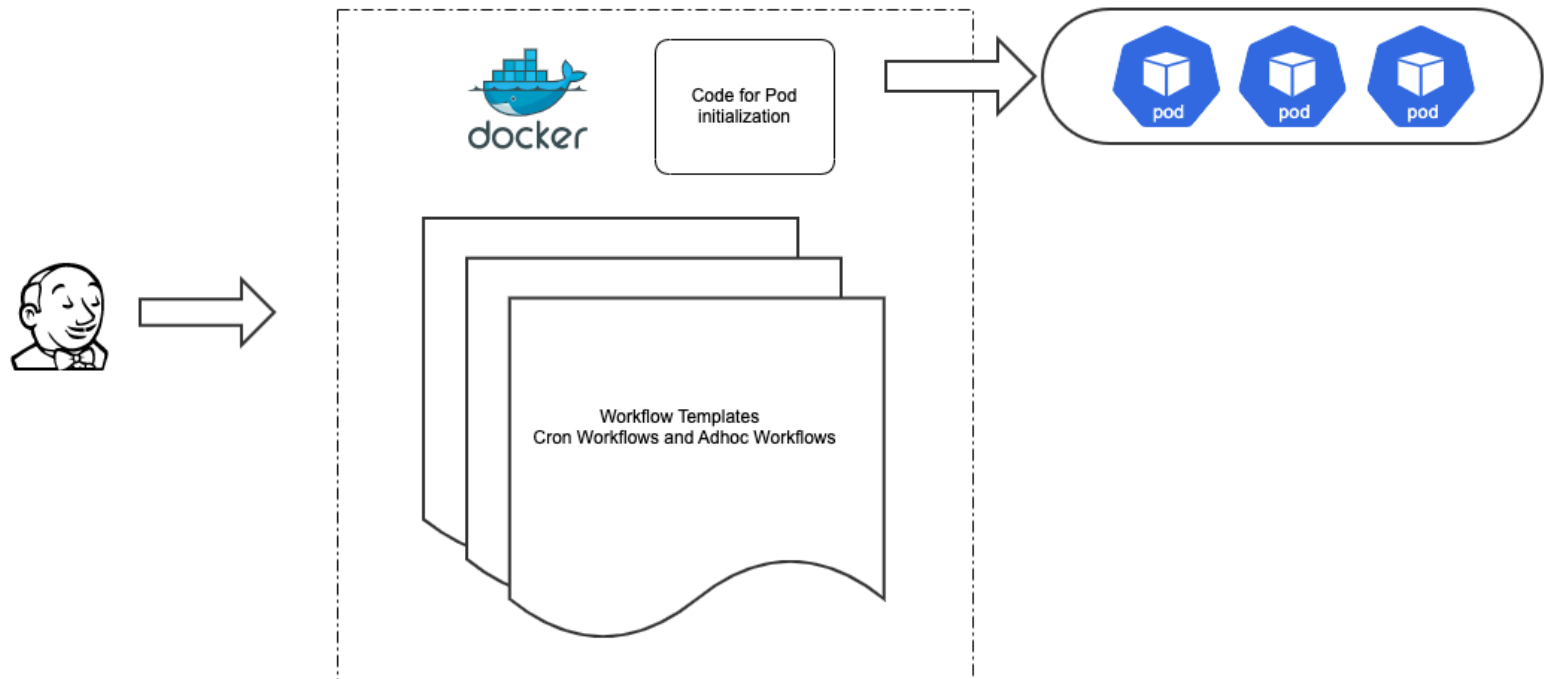
- Overview
- Prerequisites
- Data Comparison Reports Setup
- How to update/refresh Report Template
- Other DBSolo WIKI

- Overview :

Database comparison solo job has been achieved by GitOps based solution using Argo Workflows and IKS infrastructure. In this solution, we have leveraged Argo Workflows to create workflows to do our comparison. All we need is to upload the configuration files to an S3 bucket and configure the workflow parameters. Then the workflow will spin up pods on which the validation will happen, once the validation is completed, the resultant reports will be pushed to S3 bucket and the pod will be deleted.

IKS Jenkins Automation Architecture

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We have Docker Image with DBsolo installed : <https://github.intuit.com/payroll-services-platform/db-validation-image>

Code for Pod Initialisation.

```
installDBSolo() {  
    dbsolo  
    mkdir -p /usr/local/dbsolo5/results  
    cd /usr/local/dbsolo5  
    sed -i "s|-Xms128m|-Xms2128m|" commandLine  
    sed -i "s|-Xmx1024m|-Xmx134024m|" commandLine  
}
```

```

runDBSolo(){
    echo "Starting DB Validation"
    commandLine -dataCompare $FILE_NAME
    echo "Completed DB Validation"
}

runShellScript(){
    mkdir -p /u01/ogg/scripts
    cp -r /usr/local/dbsolo5 /u01/ogg/scripts
    cd /u01/ogg/scripts/dbsolo5
    mv dbsolo5/dbsolo5/* .
    sudo chmod 777 qbo_commandline_all.sh
    sudo chmod 777 qbo_commandline.sh
    ./"$CUSTOM_SHELL_SCRIPT" $cluster $v_src $v_tgt $v_pg_db $xml_file $v_days $v_hours $v_schema
}

main(){
    installDBSolo
    if [[ ! -z "$CUSTOM_SHELL_SCRIPT" ]]
    then
        runShellScript
    else
        runDBSolo
    }
}
main

```

Workflow templates and structure

Ref - <https://github.intuit.com/platformops-cloudops/dataparity-deployment/tree/master/environments/e2e-usw2-eks>

To run Database comparison primarily we have 2 templates. (Adhoc and Cron schedule)

1. Adhoc template to run for all the tables for any environment based on parameters which we pass : "\$cluster \$v_src \$v_tgt \$v_pg_db \$xml_file \$v_days \$v_hours \$v_schema" (**Use it only for adhoc purpose**)

Ref - [dbsolo-shell-script-template.yaml](#)

2. Cron schedule template for clusters - [cron-e2e.yaml](#)

Notes: (Addition of new Cluster)

- a. For new cluster we have few changes mentioned below the code.
- b. Copy the .yaml existing template and name it based on cluster - cron-c1.yaml
- c. Add a new entry in connections.xml and upload in S3
- d. Change metadata : name as per cluster name - ex- cron-workflow-c1
- e. Change in parameters value ex - Cluster, v_src, v_tgt, v_pg_db

```
apiVersion: argoproj.io/v1alpha1
kind: CronWorkflow
metadata:
  name: cron-workflow-e2e #Cluster Number ex - Cluster1#
spec:
  schedule: "0 0 * * *"
  workflowSpec:
    retryStrategy:
      retryPolicy: OnError
      limit: 1
    entrypoint: dbsoloroot
    parallelism: 1
    serviceAccountName: workflow-service-account
  templates:
    - name: dbsoloroot
      steps:
        - - name: dbsolo1
            templateRef:
              name: dbsolo-shell-script-template
              template: dbsolo
            arguments:
              parameters:
                - name: bucket_name
                  value: dbsolo
                - name: directory_name
                  value: CDC/OracleToPostgres
                - name: role_arn
                  value: arn:aws:iam::475968958217:role/dataparity-dbsolo-iam-role
                - name: cluster
                  value: E2E_C23_O2P #Cluster Number ex - Cluster1#
                - name: v_src
                  value: e2e_c23_ora #Cluster Number ex - Cluster1#
                - name: v_tgt
                  value: e2e_c23_apg #Cluster Number ex - Cluster1#
                - name: v_pg_db
                  value: eqboc23p #DB Name ex - DBNAME#
                - name: xml_file
                  value: qbo_qbodata_schemas_tables_template_1.xml
                - name: v_days
                  value: 1
                - name: v_hours
```

```
value: 2
- name: v_schema
value: BOTH
```

No labels